

Scientific Review & Synthesis

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Deposition of Particulate Matter of Different Size Fractions on Leaf Surfaces and in Waxes of Urban Forest Species

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1. Abstract & Scope

Particulate matter (PM) is a major urban pollutant. This study compared four tree species (*Acer campestre*, *Fraxinus excelsior*, *Platanus x hispanica*, *Tilia cordata*), three shrubs (*Forsythia x intermedia*, *Physocarpus opulifolius*, *Spiraea japonica*), and one climber (*Hedera helix*) commonly cultivated along streets in Warsaw, Poland. Separate gravimetric analyses quantified PM deposited on leaf surfaces (SPM) and trapped in waxes (WPM) across three size fractions (10-100 μm , 2.5-10 μm , and 0.2-2.5 μm). Significant species-specific differences in PM capture were found. Shrubs, particularly *Spiraea japonica*, accumulated the highest total PM load, whereas *Platanus* was the least effective. Waxes were shown to permanently encapsulate fine and ultrafine particles, protecting them from rain wash-off.

2. Methodology & Experimental Parameters

Sequential gravimetric washing (water followed by chloroform). Cascade vacuum filtration on Whatman filters. Normalization per unit leaf area.

3. Core Findings & Data Synthesis

Spiraea japonica was the most effective accumulator; *Platanus x hispanica* was the least. PM₁₀₋₁₀₀ represents the bulk of mass, but the fine PM_{2.5} fraction was concentrated within the epicuticular wax layer. No linear correlation existed between wax weight and total PM.

4. Strategic Relevance to JU and Kolkata Planning

Establishes the sequential gravimetric extraction method and proves that shrubs are crucial lower-strata roadside filters.

